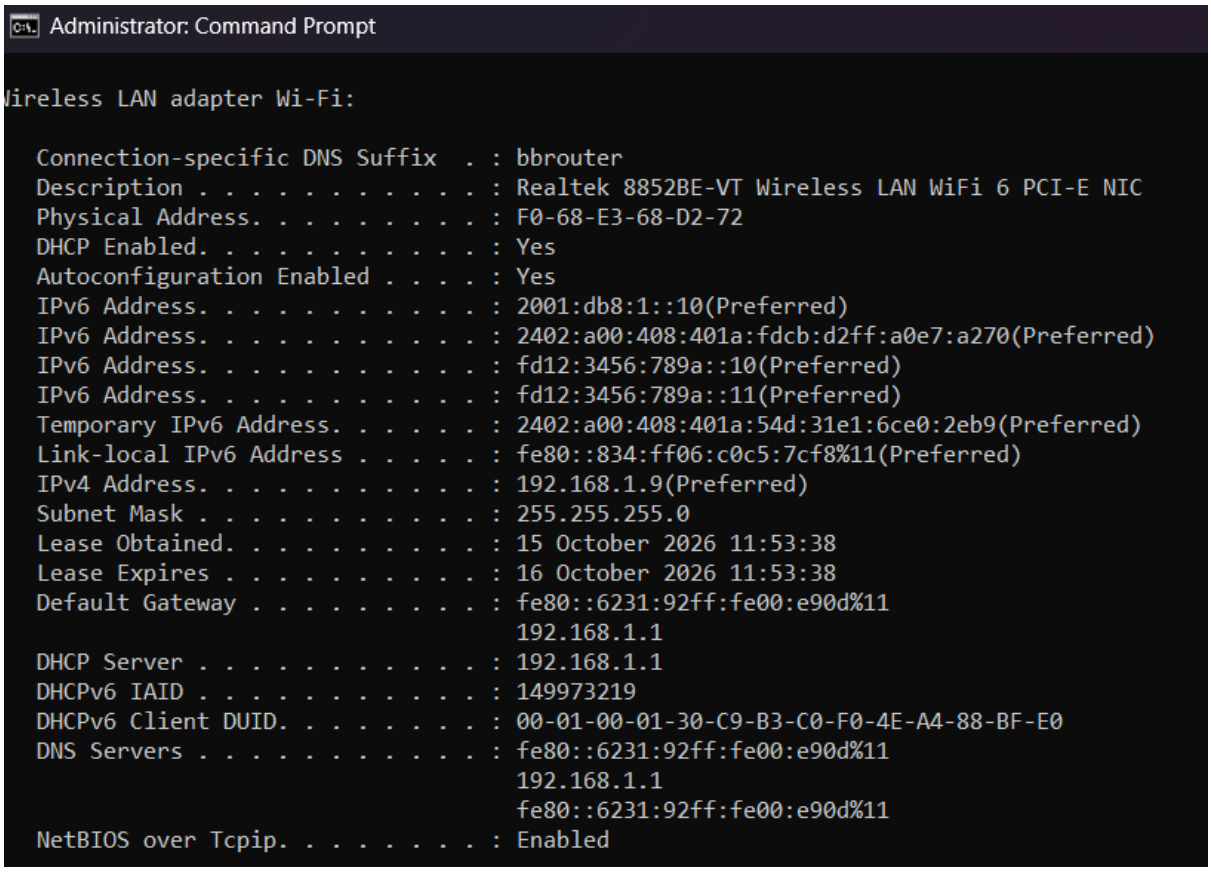
Task 1:

Use the Windows command prompt to run the command ipconfig /all on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.



Task 2:

Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

Without network:

```
C:\Windows\System32>ping www.google.com
Ping request could not find host www.google.com. Please check the name and try again.
```

With network:

```
C:\Windows\System32>ping www.google.com

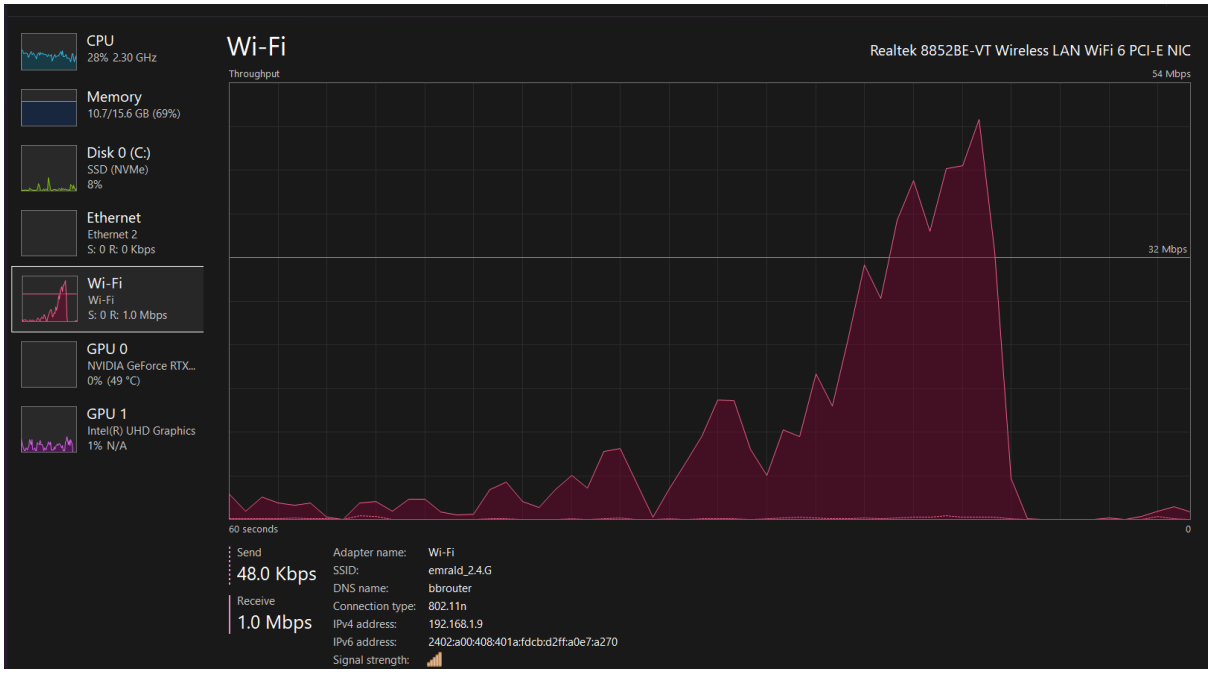
Pinging www.google.com [2001:4860:4829:7700::] with 32 bytes of data:
Reply from 2001:4860:4829:7700::: time=13ms
Reply from 2001:4860:4829:7700::: time=13ms
Reply from 2001:4860:4829:7700::: time=13ms
Reply from 2001:4860:4829:7700::: time=13ms

Ping statistics for 2001:4860:4829:7700:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 13ms, Maximum = 13ms, Average = 13ms
```

Condition	**Expected result**	**Meaning**
Wi-Fi disconnected	Request timed out, destination unreachable, or name-resolution error	Internet connectivity is unavailable; exact message depends on the system.
Wi-Fi connected	Reply from Google's server with time in milliseconds	The computer can reach Google's server.
Connected but slow	Replies with high time in milliseconds	Network response is slow.
Connected but unstable	Some replies and some timeouts	Connection may be intermittent or experiencing packet loss.

Task 3:

Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.



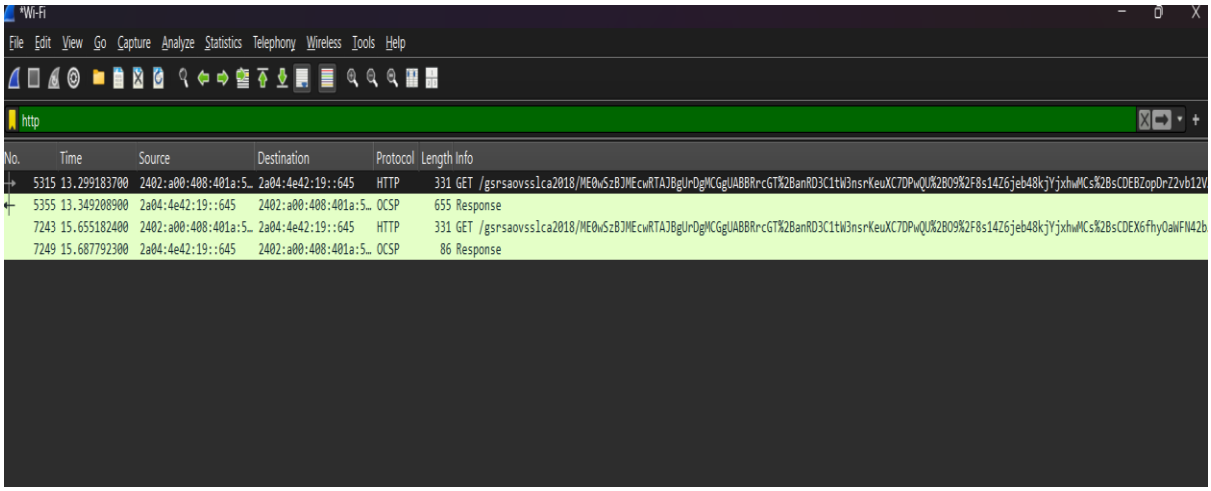
I opened Task Manager and monitored my laptop's network usage while streaming a YouTube video for five minutes. I checked the network activity in the Performance tab and identified the application using the most bandwidth. I recorded the peak network usage and observed how streaming affected my internet connection.

Task 4:

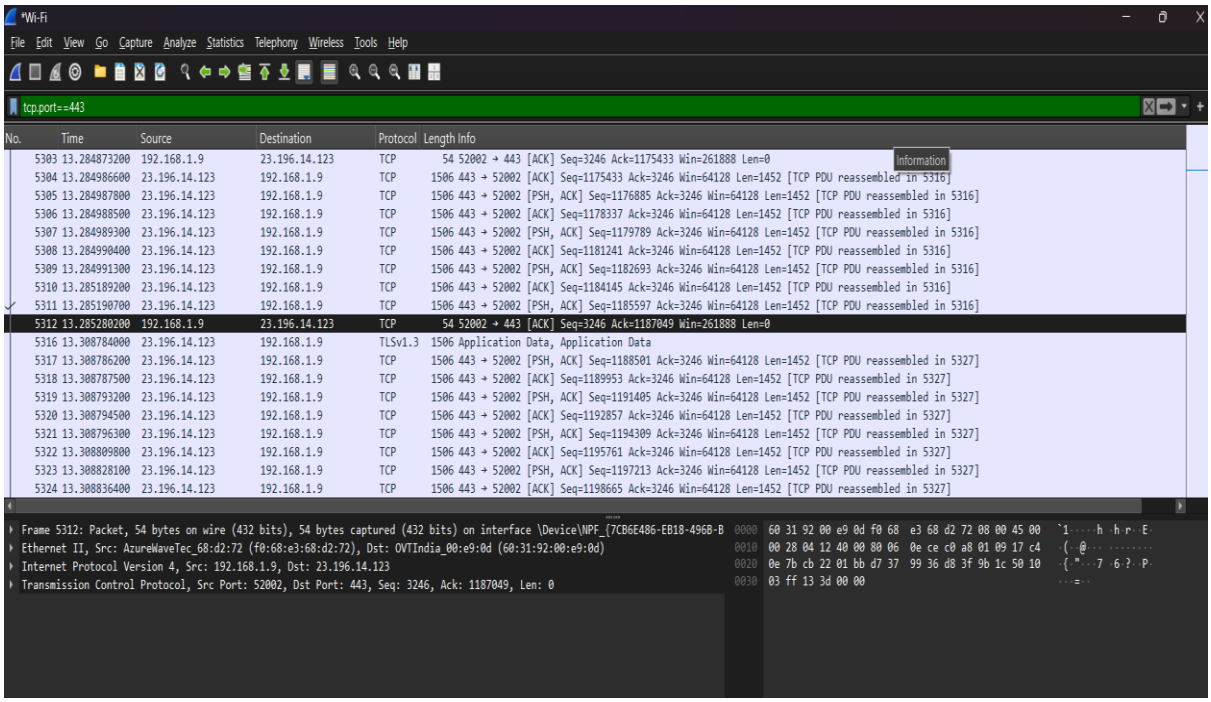
Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.

Answer:

HTTP



HTTPS



Task 5:

Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

Answer:

If my friend experiences slow internet while playing online games, I would suggest the following steps:

1. **Use an Ethernet Cable:** Connect the gaming computer directly to the router to reduce wireless interference and improve stability.
2. **Restart the Router:** Turn off the router, wait for approximately 30 seconds, and turn it on again.
3. **Stop Background Downloads:** Pause large downloads, cloud backups, and unnecessary streaming applications.
4. **Improve Wi-Fi Signal:** Place the router in a central, open location near the gaming device.
5. **Choose the Correct Wi-Fi Band:** Use the 5 GHz band when nearby for higher speeds, or 2.4 GHz when greater range is needed.
6. **Check Ping and Packet Loss:** Use the ping command to identify high response times and lost packets.
7. **Update Network Drivers:** Install appropriate network adapter driver updates.
8. **Limit Network Congestion:** Ask other users to pause heavy downloads while gaming.
9. **Check Router Firmware:** Update router firmware using the manufacturer's official instructions.
10. **Contact the ISP:** If the problem continues, ask the internet service provider to check for service issues.

Suggestion I Would Try First: I would connect the gaming computer directly to the router using an Ethernet cable because it can reduce Wi-Fi interference and improve connection stability.

Conclusion: These troubleshooting steps can help reduce lag, improve connection reliability, and provide a better online gaming experience.